Depreciation Accounting Under Ind AS 16

Executive Overview: Ind AS 16 (Property, Plant and Equipment) governs the accounting treatment for tangible fixed assets, including their cost measurement, subsequent recognition, and systematic depreciation. This study note synthesises the definitions, characteristics, causes, and calculation methodologies required under the standard.

Meaning and Core Concept of Depreciation

Under Ind AS 16, depreciation is defined as the systematic allocation of the depreciable amount of an asset over its estimated useful life. It represents the gradual, non-cash reduction in the carrying value of a fixed asset resulting from continuous use, the passage of time, or technological obsolescence.

As a primary accounting rule, depreciation is recognised as an expense in the Statement of Profit and Loss, except in specific circumstances where it is permitted to be capitalized into the carrying amount of another asset (for instance, equipment used in constructing a new internal facility).

Depreciable Amount = Cost of Asset − Residual Value

Nature & Characteristics of Depreciation

Depreciation is distinct from other business expenditures. Under Ind AS 16, its key operational and accounting characteristics include:

- **It is a non-cash expense:** Depreciation does not involve an actual cash outflow in the period in which it is charged.
- **It is a systematic allocation:** It is not an arbitrary reduction in market value. The depreciable amount is allocated over its useful life using a rational and consistent methodology.
- **It relates to tangible fixed assets:** It generally applies to depreciable tangible physical assets such as buildings, machinery, office furniture, and vehicles.
- **It is based on useful life:** The charge in each period depends on the estimated economic useful life of the asset to the entity.

- **It reduces the carrying amount:** Depreciation directly reduces the book value (carrying amount) of the asset on the balance sheet over time.
- **It is an expense:** Unless capitalized into the cost of another asset, it is recognized directly as an operating expense in the Statement of Profit and Loss.
- **It is an allocation process, not a valuation process:** Its primary focus is systematically allocating cost, rather than determining or reflecting the current fair market value of the asset.

Causes of Depreciation

Assets lose their utility and value due to several physical, environmental, and economic factors. The primary causes under standard asset accounting include:

- **Wear and Tear:** Deterioration caused by continuous physical use. For example, industrial machinery operated daily gradually loses operational efficiency.
- **Passage of Time:** Certain assets lose utility simply due to the passage of time, irrespective of active usage (e.g., leasehold improvements or legal rights with strict expiry terms).
- **Obsolescence:** Technological advancements or changes in production methods can make assets outdated. For example, computer hardware quickly becomes obsolete when more efficient models emerge.
- **Depletion:** The physical reduction of natural resources through extraction or consumption, applicable specifically to wasting assets like mines, oil wells, and quarries.
- **Physical Deterioration:** Damage resulting from environmental exposure, such as rust, corrosion, decay, moisture, and extreme heat, which naturally shorten an asset's useful life.
- **Legal or Contractual Limits:** The legal or contractual framework governing an asset may restrict its utility (e.g., assets acquired under lease agreements with fixed contract terms).

Factors Affecting Depreciation under Ind AS 16

To calculate depreciation accurately, an entity must evaluate several variables and accounting estimates. Ind AS 16 outlines seven critical factors that influence the depreciation charge:

1. Cost of the Asset

Includes the purchase price, import duties, non-refundable taxes, and any directly attributable costs to bring the asset to its location and working condition for its intended use.

2. Residual Value

The estimated amount an entity would currently obtain from disposing of the asset, after deducting disposal costs, assuming it is already of the age/condition expected at the end of its useful life. A higher residual value results in a lower depreciable amount.

3. Useful Life

The period over which the asset is expected to be available for use, or the number of production units expected from it. A longer useful life reduces the annual depreciation expense.

4. Method of Depreciation

The selected method must reflect the pattern in which the asset's future economic benefits are

expected to be consumed (e.g., Straight-Line Method, Written Down Value Method, or Units of Production Method).

5. Pattern of Consumption

The expected consumption of economic benefits directly dictates the choice of depreciation method, aligning the expense recognition with revenue-generating utility.

6. Significant Parts of an Asset

Under the component accounting approach, if an asset contains significant parts with different useful lives (e.g., an aircraft frame vs. its engines), each part must be depreciated separately.

7. Review of Useful Life & Residual Value

Residual values and useful lives must be reviewed at least at each financial year-end. If expectations change, the adjustment is treated prospectively as a change in accounting estimate under Ind AS 8.

Practical Illustration and Formula

The following practical case study demonstrates the calculation of depreciation under the Straight-Line Method (SLM) based on the principles outlined above.

Parameter	Value (₹)	Description / Formula
Initial Cost of Machine	10,00,000	Purchase price and direct installation costs
Estimated Residual Value	1,00,000	Disposal value net of costs at end of life
Depreciable Amount	9,00,000	Cost of Asset (₹10,00,000) − Residual Value (₹1,00,000)
Estimated Useful Life	9 Years	Expected period of productive utility
Annual Depreciation (SLM)	**1,00,000**	**Depreciable Amount (₹9,00,000) ÷ Useful Life (9 Years)**

DIFFERENCES

Basis	Depreciation	Depletion	Amortisation	Dilapidation
Meaning	Reduction/allocation of cost of tangible assets	Reduction of natural resources through extraction	Allocation of cost of intangible assets	Loss/damage to property requiring restoration
Applies to	Machinery, buildings, vehicles	Mines, oil wells, quarries	Patents, copyrights, licences	Leased/rented properties
Example	Depreciation of machinery	Coal extracted from a mine	Patent amortised over its life	Restoration of a damaged building
Key idea	**Tangible assets**	**Natural resources**	**Intangible assets**	**Property damage/restoration**